

ST AUGUSTINE PIER, FL, USA

WMO: 994410

Lat: 29.8573N

Lon: 81.2636W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.4	(c) 4.6	(d) -8.4	(e) 1.8	(f) 4.3	(g) -5.2	(h) 2.4	(i) 6.9	(j) 16.7	(k) 11.8	(l) 14.8	(m) 11.5	(n) 5.3	(o) 320	(p) 0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.9	(c) 32.1	(d) 24.5	(e) 30.7	(f) 24.5	(g) 29.7	(h) 24.7	(i) 26.8	(j) 28.4	(k) 26.3	(l) 28.2	(m) 25.9	(n) 27.9	(o) 4.4	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.8	27.7	25.8	21.1	27.4	25.3	20.5	27.2	83.6	28.4	81.6	28.2	79.9	28.1	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	12.9	11.0	-0.7	34.4	2.4	1.5	-2.5	35.5	-3.9	36.4	-5.3	37.3	-7.0	38.4
			-2.6	27.2	1.9	1.0	-4.0	27.9	-5.1	28.5	-6.2	29.1	-7.6	29.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	13.3	14.9	16.9	20.1	23.5	25.7	26.5	26.8	26.0	23.1	18.6	15.2
	DBStd	5.69	4.45	4.11	3.74	2.83	2.09	1.52	1.46	1.37	1.41	2.99	3.77	4.44
	HDD10.0	43	21	8	3	0	0	0	0	0	0	0	1	10
	HDD18.3	515	162	108	69	16	1	0	0	0	0	6	42	113
	CDD10.0	4027	123	144	216	303	418	472	513	520	479	406	260	173
	CDD18.3	1457	6	11	24	68	161	222	254	262	229	153	51	17
	CDH23.3	10806	13	36	71	159	779	1735	2275	2552	2116	937	110	24
CDH26.7	2088	1	3	11	34	118	304	534	626	377	75	6	1	
Wind	WSAvg	5.1	5.0	5.2	5.3	5.2	5.5	4.6	4.0	4.1	5.6	5.8	5.5	5.0
Precipitation	PrecAvg	1292	72	72	87	63	75	174	138	163	219	122	56	64
	PrecMax	1682	201	217	225	224	180	284	292	327	433	306	144	257
	PrecMin	835	24	16	4	14	26	18	42	73	51	8	4	10
	PrecStd	225	47	52	60	53	48	67	59	65	113	78	36	64
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	27.1	28.1	29.9	32.0	34.2	33.7	34.0	32.0	29.9	27.5	26.1
		MCWB	19.2	20.7	19.5	20.6	21.9	25.7	25.6	24.4	23.1	23.5	23.4	22.0
	2%	DB	22.2	24.1	25.4	27.3	29.9	31.1	31.7	31.5	30.1	28.3	25.6	23.5
		MCWB	18.0	18.6	18.9	20.3	21.8	24.5	25.2	25.3	23.6	23.7	23.0	20.1
	5%	DB	20.3	21.4	22.9	24.8	27.7	29.4	30.2	30.2	28.9	27.4	24.7	21.9
		MCWB	17.8	17.7	18.8	20.5	22.3	24.4	25.2	25.6	24.5	23.7	22.5	19.6
	10%	DB	19.0	19.9	21.3	23.6	26.2	28.1	29.0	29.3	28.3	26.8	23.6	20.8
		MCWB	17.3	17.6	18.4	20.4	22.6	24.5	25.2	25.6	24.5	23.3	20.8	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	21.3	21.5	23.3	25.0	26.6	28.4	27.7	26.7	26.6	25.7	21.8
		MCDB	22.9	24.0	24.4	24.7	26.8	28.8	28.5	29.7	28.1	27.0	26.0	23.6
	2%	WB	19.2	19.5	20.4	22.4	24.1	25.8	26.8	26.8	26.1	25.6	24.8	20.8
		MCDB	21.3	21.4	22.7	23.8	25.8	28.0	28.2	28.8	27.8	27.0	25.0	22.1
	5%	WB	18.5	18.4	19.6	21.8	23.7	25.4	26.1	26.4	25.6	24.7	23.0	20.2
		MCDB	19.9	19.8	21.5	23.1	25.6	27.6	28.3	28.3	27.6	26.6	24.2	21.3
	10%	WB	17.5	17.0	19.0	21.1	23.2	25.0	25.6	26.0	25.2	23.9	21.5	19.3
		MCDB	18.7	18.8	20.7	22.8	25.3	27.2	27.9	28.1	27.4	26.0	23.2	20.4
Mean Daily Temperature Range	5% DB	MDBR	7.0	6.5	6.1	5.6	4.9	5.1	5.2	4.9	4.1	4.6	5.9	6.4
		MCDBR	8.5	9.1	9.0	8.2	8.6	7.7	7.5	7.2	5.6	4.9	6.7	7.0
		MCWBR	4.8	6.2	4.8	4.0	3.5	3.4	3.5	3.1	2.4	2.7	4.2	5.0
	5% WB	MCDBR	6.1	7.8	6.1	5.3	5.0	5.4	5.5	5.1	3.8	3.9	3.4	5.1
		MCWBR	4.3	6.3	4.3	3.6	3.0	3.1	3.3	2.9	2.4	2.8	2.8	4.3
Clear-Sky Solar Irradiance	taub	0.339	0.352	0.371	0.396	0.433	0.469	0.495	0.480	0.427	0.399	0.362	0.347	
	taud	2.541	2.513	2.462	2.373	2.317	2.279	2.212	2.260	2.403	2.448	2.499	2.543	
	Ebn at Noon	897	915	916	899	862	826	804	813	849	852	862	870	
	Edh at Noon	87	97	109	123	131	135	144	136	114	103	90	83	
All-Sky Solar Radiation	RadAvg	3.17	3.81	4.98	6.08	6.44	6.00	5.95	5.58	4.82	4.25	3.42	2.88	
	RadStd	0.23	0.38	0.43	0.36	0.41	0.48	0.36	0.37	0.39	0.34	0.20	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		+0.44	+0.77	N/A	N/A	N/A	N/A	-9	-62	+160	+116

Nomenclature: See separate page